Lecture 40

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1 Orthogonality and Coordinate Systems

1.1 Orthogonal Vectors

Two vectors a and b are called **orthogonal** (perpendicular) if the angle φ between them is $\frac{\pi}{2}$ ($90°$).

- Notation: $a \perp b$

1.2 Rectangular Cartesian Coordinate System

A rectangular Cartesian coordinate system is a set of points and unit vectors mutually orthogonal to each other.

1.2.1 In $\mathbb{R}^2$

- Vectors: e_1, e_2
- Properties:
  - $|e_1| = 1, |e_2| = 1$
  - $\langle e_1, e_2 \rangle = 0$ (Orthogonal)
- Vector representation: $a = (a_1, a_2) = a_1 e_1 + a_2 e_2$

1.2.2 Distance between two points in $\mathbb{R}^2$

Given points $A(x_1, y_1)$ and $B(x_2, y_2)$:

$$|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

1.2.3 In $\mathbb{R}^3$

- Defined by origin O and three mutually orthogonal unit vectors e_1, e_2, e_3.
- Given points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$:

$$|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2 Scalar Product (Inner Product)

2.1 Definition

The scalar (or dot/inner) product of two vectors a and b is the number:

$$\langle a, b\rangle = |a| \cdot |b| \cdot \cos\varphi$$

where φ is the angle between a and b.

2.2 Geometric Interpretation

- If the angle is **acute** ($0 \leq \varphi < 90°$): $\langle a, b\rangle > 0$
- If the angle is **obtuse** ($90° < \varphi \leq 180°$): $\langle a, b\rangle < 0$
- If vectors are orthogonal: $\langle a, b\rangle = 0$

2.3 Properties

1. $\langle a, b\rangle = \langle b, a\rangle$ (Symmetry)
2. $\langle \lambda a, b\rangle = \lambda\langle a, b\rangle$ for $\lambda \in \mathbb{R}$ (Linearity)
3. $\langle a + b, c\rangle = \langle a, c\rangle + \langle b, c\rangle$ (Distributivity)
4. $\langle a, a\rangle = 0 \iff a = 0$ (Positive definiteness)

3 Inner Product in Coordinates

3.1 Derivation in $\mathbb{R}^2$

Let $a = a_1e_1 + a_2e_2$ and $b = b_1e_1 + b_2e_2$. Using the properties of the inner product and the fact that coordinates are orthonormal ($\langle e_i, e_j\rangle = \delta_{ij}$):

$$\begin{aligned}
\langle a, b\rangle &= \langle a_1e_1 + a_2e_2, b_1e_1 + b_2e_2\rangle \\
&= a_1b_1\langle e_1, e_1\rangle + a_1b_2\langle e_1, e_2\rangle + a_2b_1\langle e_2, e_1\rangle + a_2b_2\langle e_2, e_2\rangle \\
&= a_1b_1(1) + a_1b_2(0) + a_2b_1(0) + a_2b_2(1) \\
&= a_1b_1 + a_2b_2
\end{aligned}$$

3.2 Formula for Angle

Since $\langle a, b\rangle = |a| \cdot |b| \cdot \cos\varphi = a_1b_1 + a_2b_2$:

$$\cos\varphi = \frac{a_1b_1 + a_2b_2}{\sqrt{a_1^2 + a_2^2} \cdot \sqrt{b_1^2 + b_2^2}}$$

3.3 Generalization to $\mathbb{R}^3$

For $a = (a_1, a_2, a_3)$ and $b = (b_1, b_2, b_3)$:

$$\langle a, b \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\cos\varphi = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

4 Equation of a Line

4.1 Ways to Define a Line

1. Two points.
2. A point and a direction vector.
3. A point and an orthogonal (normal) vector.

4.2 Normal Form (Point and Orthogonal Vector)

Given a point $P_0(x_0, y_0)$ and a normal vector $n = (a, b)$. For any point $P(x, y)$ on the line, the vector $\vec{P_0P}$ is orthogonal to n:

$$\langle n, \vec{P_0P} \rangle = 0$$

$$a(x - x_0) + b(y - y_0) = 0$$

Let $c = -(ax_0 + by_0)$, then the **general equation of a line** is:

$$ax + by + c = 0$$

where $n = (a, b)$ is the normal vector.

4.3 Angle Between Two Lines

The angle φ between two lines l_1 and l_2 is the angle between their normal vectors n_1 and n_2. Given lines:

- $l_1 : a_1 x + b_1 y + c_1 = 0 \implies n_1 = (a_1, b_1)$
- $l_2 : a_2 x + b_2 y + c_2 = 0 \implies n_2 = (a_2, b_2)$

$$\cos\varphi = \frac{|\langle n_1, n_2 \rangle|}{|n_1| \cdot |n_2|} = \frac{|a_1 a_2 + b_1 b_2|}{\sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2}}$$

4.4 Two-Point Form

Equation of the line passing through $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$. A point $P(x, y)$ is on the line if $\vec{P_1P}$ is parallel to $\vec{P_1P_2}$.

$$\vec{P_1P} = \lambda \vec{P_1P_2}$$

In coordinates:

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1}$$

5 Equation of a Plane in $\mathbb{R}^3$

5.1 Ways to Define a Plane

1. Three points.
2. A point and a normal vector.
3. Two lines.

5.2 Normal Form

Given a point $P_0(x_0, y_0, z_0)$ and a normal vector $n = (a, b, c)$. For any point $P(x, y, z)$ on the plane:

$$\langle n, \vec{P_0P} \rangle = 0$$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Let $d = -(ax_0 + by_0 + cz_0)$, then the general equation is:

$$ax + by + cz + d = 0$$

5.3 Angle Between Two Planes

The angle between two planes is the angle between their normal vectors. Given planes with normals $n_1 = (a_1, b_1, c_1)$ and $n_2 = (a_2, b_2, c_2)$:

$$\cos\varphi = \frac{\langle n_1, n_2 \rangle}{|n_1| \cdot |n_2|} = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$